Assuming excluded middle every cardinal number has a strictly greater one

The Univalent Foundations Program

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Editorial note. The counted statement is the author's unlabelled cor environment at pinned-original-source.tex lines 867-869, i.e. the corollary of section 10.2 that follows Cantor's theorem; the proof environment opens at line 870, so the pairing is by adjacency.

Editorial note. Numbering: this corollary carries no label, so its number is part of its identity. The retained counter values (chapter 10, section 2, theorem offset 8) were chosen so that the context environments reproduce their first-edition numbers and the counted corollary receives the next free one; the resulting sequence is 10.2.9, 10.2.10, 10.2.11, 10.2.12 for the four contexts and 10.2.13 for the counted corollary.

Editorial note. The four contexts are retained because the counted proof cites Lemma 10.2.9 by label and invokes Cantor's theorem in prose; keeping the three intervening statements of section 10.2 also makes the printed numbering of the card agree with the first-edition numbering, since those environments consume the numbers between 10.2.9 and 10.2.13.

Editorial note. The numbers 10.2.9 for thm:injsurj and 10.2.6 for card:exp are taken from the first-edition label-number dump; the intervening environments were counted against those two anchors, and the print of thm:injsurj in the assembled card can be read in the auxiliary file as an independent check.

Editorial note. No retained line refers to a label outside the retained material, so the package declares no external reference.

Editorial scope. One original result of the pinned first-edition chapter, counted once: the existence of a strictly greater cardinal for every cardinal under excluded middle (unlabelled corollary, original Corollary 10.2.13), with the statements it cites retained as context.

Allowed standard prerequisites (editorial).

- Cardinal numbers and cardinal inequality. card is the set-truncation of the type of sets; alpha<=beta means there is an injection from a representative of alpha to a representative of beta.. It is the relation the statement is about, and the proof reasons on representatives.
- Cantor's theorem. For a set A there is no surjection A -> (A->bool).. The second half of the proof contradicts it.

- Injections versus surjections. Lemma 10.2.9: under excluded middle an injection A->B into an inhabited B yields a surjection A->B.. It turns the assumed injection (A->bool) -> A into a surjection, which Cantor's theorem forbids.

Lemma 10.2.9. *Consider the following statements:*

(i) There is an injection $A \to B$.
(ii) There is a surjection $B \to A$.

Then, assuming excluded middle:

- *Given $a_0 : A$, we have (i)→(ii).*
- *Therefore, if A is merely inhabited, we have (i) → merely (ii).*
- *Assuming the axiom of choice, we have (ii) → merely (i).*

Theorem 10.2.10 (Schroeder–Bernstein)**.** *Assuming excluded middle, for sets A and B we have*

$$\mathsf{inj}(A, B) \to \mathsf{inj}(B, A) \to (A \cong B)$$

Corollary 10.2.11. *Assuming excluded middle, cardinal inequality is a partial order, i.e. for $\alpha, \beta : \mathsf{Card}$ we have*

$$(\alpha \leq \beta) \to (\beta \leq \alpha) \to (\alpha = \beta).$$

Theorem 10.2.12 (Cantor)**.** *For $A : \mathsf{Set}$, there is no surjection $A \to (A \to \mathbf{2})$.*

Corollary 10.2.13. *Assuming excluded middle, for any $\alpha : \mathsf{Card}$, there is a cardinal β such that $\alpha \leq \beta$ and $\alpha \neq \beta$.*

Proof. Let $\beta = 2^\alpha$. Now we want to show a mere proposition, so by induction we may assume α is $|A|_0$, so that $\beta \equiv |A \to \mathbf{2}|_0$. Using excluded middle, we have a function $f : A \to (A \to \mathbf{2})$ defined by

$$f(a)(a') :\equiv \begin{cases} 1_{\mathbf{2}} & a = a' \\ 0_{\mathbf{2}} & a \neq a'. \end{cases}$$

And if $f(a) = f(a')$, then $f(a')(a) = f(a)(a) = 1_{\mathbf{2}}$, so $a = a'$; hence f is injective. Thus, $\alpha \equiv |A|_0 \leq |A \to \mathbf{2}|_0 \equiv 2^\alpha$.

On the other hand, if $2^\alpha \leq \alpha$, then we would have an injection $(A \to \mathbf{2}) \to A$. By Lemma 10.2.9, since we have $(\lambda x.\, 0_{\mathbf{2}}) : A \to \mathbf{2}$ and excluded middle, there would then be a surjection $A \to (A \to \mathbf{2})$, contradicting Cantor's theorem. □